Candidate Intelligence Engine

A Multi-source Candidate Data Extraction & Transformation Pipeline

The Candidate Intelligence Engine is a state-of-the-art data pipeline designed to ingest candidate profiles from diverse sources, validate and merge them natively, and project them into a highly reliable canonical JSON schema.

With advanced text parsing and an intuitive interactive dashboard, this engine bridges the gap between messy source data (PDF resumes, CSVs, JSON) and downstream HR systems.

🌟 Key Features

- **Multi-Source Ingestion**: Process unstructured resumes (PDF), structured recruiter exports (CSV, Notes), and API payloads (ATS/GitHub JSON).
- **Advanced Document Parsing**: The pipeline extracts visible text AND explicitly uncovers hidden hyperlink annotations within PDFs to ensure URLs (like portfolios) are never missed.
- **Intelligent Prioritization**: Smart heuristics actively filter out academic email domains or generic tech links, proactively prioritizing primary contact methods and actual portfolio links (like `.github.io`).
- **Canonical JSON Assurance**: Deduplicates parsed items (like emails/phones) and rigorously projects merged data into a fixed, canonical schema.
- **Dynamic Visualization Dashboard**:
  - **Interactive Mode**: A gorgeous graphical UI dynamically renders timelines, skill tags, and confidence scores.
  - **Textual Mode**: A beautifully syntax-highlighted code block exposing the underlying Canonical JSON schema in real time.
  - **Quick Actions**: Instantly download the output as a `.json` file or copy it to the clipboard!

🚀 Running the Engine

This system runs purely on Python. No Docker, Kubernetes, or complex DevOps pipelines are required—making it highly accessible for local evaluation!

1. Install Dependencies

Ensure you have Python 3.10+ installed. Open your command prompt and run: `pip install -r requirements.txt`

2. Launch the Application

Start the unified backend API and frontend dashboard: `python -m uvicorn candidate_transformer.api.app:create_app --factory`

3. Open the Dashboard

Navigate your web browser to: `http://127.0.0.1:8000`

🏗 System Architecture Flow

To make this understandable for everyone, here is how data flows through the system:

1. **Input Stage**: You upload a Resume PDF (or other files) into the web dashboard.
2. **Extraction Layer**: The engine reads the PDF, extracts text, and digs into the file metadata to find hidden links.
3. **Merging Layer**: If you upload multiple files for the same person (e.g., a Resume and a GitHub Profile), the engine securely merges them into one single profile, removing any duplicate information.
4. **Projection Layer**: The merged data is formatted into a strict, standardized JSON format that HR software can understand.
5. **Output Stage**: The dashboard displays the results beautifully on your screen.

📁 Code Organization

The codebase is organized into cleanly separated domains to ensure scalability and maintainability.

- **src/api/** - FastAPI web server and static frontend dashboard assets (HTML/CSS/JS).

- **src/extraction/** - The intelligence layer containing the advanced parsers (PDF, CSV, JSON) and regex heuristics.
- **src/merge/** - The logic engine that deduplicates and merges partial candidates into a unified record.
- **src/projection/** - The output layer that maps the merged record into the final Canonical JSON schema.
- **schemas/** - JSON Schema definitions ensuring output compliance.
- **tests/** - Comprehensive unit and integration test suite.